

Experiment: Sentiment Analysis using Hugging Face Pipeline

Objective

Demonstrate how a pretrained Transformer model predicts the sentiment of text.

Concepts Covered

- ✓ Transformers
- ✓ Hugging Face Pipeline
- ✓ Pretrained Models
- ✓ Positive/Negative Sentiment
- ✓ Confidence Scores

Install Libraries

```
pip install transformers torch
```

Python Program

```
"""
```

```
Experiment 1
```

```
Sentiment Analysis using Hugging Face Pipeline
```

```
"""
```

```
from transformers import pipeline
```

```
# Load pretrained sentiment analysis pipeline
```

```
classifier = pipeline("sentiment-analysis")
```

```
print("=" * 60)
```

```
print("Hugging Face Sentiment Analysis Demo")
```

```
print("=" * 60)
```

```
# Example sentences
```

```
texts = [
```

```
    "I absolutely love this course.",
```

```
    "The movie was terrible and boring.",
```

```
    "The food was delicious.",
```

```
    "The service was very disappointing.",
```

```
    "Artificial Intelligence is changing the world.",
```

```
    "I am very excited about learning Transformers."
```

```
]
```

```
# Predict sentiments
```

```
results = classifier(texts)
```

```
print("\nResults:\n")
```

for sentence, result in zip(texts, results):

```
print("Sentence :", sentence)
print("Sentiment:", result["label"])
print("Confidence:", round(result["score"],4))
print("-"*60)
```

Interactive prediction

while True:

```
text = input("\nEnter your own sentence (or quit): ")
```

```
if text.lower()=="quit":
    break
```

```
prediction = classifier(text)[0]
```

```
print("\nPrediction")
```

```
print("Label :", prediction["label"])
print("Score :", round(prediction["score"],4))
```

Sample Output

Sentence : I absolutely love this course.

Sentiment : POSITIVE

Confidence : 0.9998

Sentence : The movie was terrible and boring.

Sentiment : NEGATIVE

Confidence : 0.9995

Learning Outcomes

Students learn

- a) Loading pretrained models
- b) Running inference
- c) Confidence scores
- d) No training required

Experiment 2: Text Classification using Zero-Shot Classification

This is one of the most impressive Hugging Face demos.
Students can classify text into **their own labels** without training.

Install

```
pip install transformers torch
```

Python Program

```
"""
Experiment 2
Zero-Shot Text Classification
"""

from transformers import pipeline

print("="*60)
print("Zero Shot Text Classification")
print("="*60)

classifier = pipeline(
    "zero-shot-classification",
    model="facebook/bart-large-mnli"
)

text = """
Deep learning is becoming an important technology
for medical diagnosis and disease prediction.
"""

candidate_labels = [
    "Technology",
    "Health",
    "Sports",
    "Politics",
    "Education"
]

result = classifier(text, candidate_labels)

print("\nText:\n")
print(text)
```

```
print("\nClassification Results\n")

for label, score in zip(result["labels"], result["scores"]):

    print(f"{label:15s} : {score:.4f}")

print("\nPredicted Category:", result["labels"][0])

# -----
# Interactive Example
# -----

while True:

    print("\nType quit to exit")

    user_text = input("\nEnter text: ")

    if user_text.lower()=="quit":
        break

    labels = input(
        "Enter comma-separated labels:\n"
    ).split(",")

    labels = [l.strip() for l in labels]

    prediction = classifier(user_text, labels)

    print("\nPrediction\n")

    for l,s in zip(prediction["labels"],
        prediction["scores"]):

        print(f"{l:15s} {s:.4f}")

    print("\nBest Label:", prediction["labels"][0])
```

Example Output

```
Technology  0.81
Health      0.17
Education   0.01
Sports      0.00
Politics    0.00
```

Explanation for Students

Input

Deep learning helps doctors detect cancer.

Candidate labels

Technology

Health

Sports

Politics

Output

Health 0.83

Technology 0.16

Sports 0.01

Politics 0.00

The model understands language without being specifically trained for those labels.

Expected Learning Outcomes

After completing these experiments, students will be able to:

- ✓ Use pretrained Transformer models from Hugging Face.
- ✓ Perform sentiment analysis without training a model.
- ✓ Classify text using zero-shot learning with custom labels.
- ✓ Interpret prediction labels and confidence scores.
- ✓ Understand transfer learning and inference workflows in modern NLP.
- ✓ Build the foundation for advanced tasks such as question answering, summarization, named entity recognition, and chatbot development.